

## Sampling

For reviewing statistical inference, we focus on the simplest possible setting. Let  $Y$  be a random variable representing a population with a probability density function  $f(y;\theta)$ , which depends on the single parameter  $\theta$ . The probability density function (pdf) of  $Y$  is assumed to be known except for the value of  $\theta$ ; different values of  $\theta$  imply different population distributions, and therefore we are interested in the value of  $\theta$ . If we can obtain certain kinds of samples from the population, then we can learn something about  $\theta$ . The easiest sampling scheme to deal with is random sampling.

**Random Sampling.** If  $Y_1, Y_2, \dots, Y_n$  are independent random variables with a common probability density function  $f(y;\theta)$ , then  $\{Y_1, \dots, Y_n\}$  is said to be a **random sample** from  $f(y;\theta)$  [or a random sample from the population represented by  $f(y;\theta)$ ].

When  $\{Y_1, \dots, Y_n\}$  is a random sample from the density  $f(y;\theta)$ , we also say that the  $Y_i$  are *independent, identically distributed* (or *i.i.d.*) random variables from  $f(y;\theta)$ . In some cases, we will not need to entirely specify what the common distribution is.

The random nature of  $Y_1, Y_2, \dots, Y_n$  in the definition of random sampling reflects the fact that many different outcomes are possible before the sampling is actually carried out. For example, if family income is obtained for a sample of  $n = 100$  families in the United States, the incomes we observe will usually differ for each different sample of 100 families. Once a sample is obtained, we have a set of numbers, say,  $\{y_1, y_2, \dots, y_n\}$ , which constitute the data that we work with. Whether or not it is appropriate to assume the sample came from a random sampling scheme requires knowledge about the actual sampling process.

Random samples from a Bernoulli distribution are often used to illustrate statistical concepts, and they also arise in empirical applications. If  $Y_1, Y_2, \dots, Y_n$  are independent random variables and each is distributed as Bernoulli( $\theta$ ), so that  $P(Y_i = 1) = \theta$  and  $P(Y_i = 0) = 1 - \theta$ , then  $\{Y_1, Y_2, \dots, Y_n\}$  constitutes a random sample from the Bernoulli( $\theta$ ) distribution. As an illustration, consider the airline reservation example carried along in Appendix B. Each  $Y_i$  denotes whether customer  $i$  shows up for his or her reservation;  $Y_i = 1$  if passenger  $i$  shows up, and  $Y_i = 0$  otherwise. Here,  $\theta$  is the probability that a randomly drawn person from the population of all people who make airline reservations shows up for his or her reservation.

For many other applications, random samples can be assumed to be drawn from a normal distribution. If  $\{Y_1, \dots, Y_n\}$  is a random sample from the Normal( $\mu, \sigma^2$ ) population, then the population is characterized by two parameters, the mean  $\mu$  and the variance  $\sigma^2$ . Primary interest usually lies in  $\mu$ , but  $\sigma^2$  is of interest in its own right because making inferences about  $\mu$  often requires learning about  $\sigma^2$ .

## C.2 Finite Sample Properties of Estimators

In this section, we study what are called finite sample properties of estimators. The term “finite sample” comes from the fact that the properties hold for a sample of any size, no matter how large or small. Sometimes, these are called small sample properties. In Section C.3, we cover “asymptotic properties,” which have to do with the behavior of estimators as the sample size grows without bound.

## Estimators and Estimates

To study properties of estimators, we must define what we mean by an estimator. Given a random sample  $\{Y_1, Y_2, \dots, Y_n\}$  drawn from a population distribution that depends on an unknown parameter  $\theta$ , an **estimator** of  $\theta$  is a rule that assigns each possible outcome of the sample a value of  $\theta$ . The rule is specified before any sampling is carried out; in particular, the rule is the same regardless of the data actually obtained.

As an example of an estimator, let  $\{Y_1, \dots, Y_n\}$  be a random sample from a population with mean  $\mu$ . A natural estimator of  $\mu$  is the average of the random sample:

$$\bar{Y} = n^{-1} \sum_{i=1}^n Y_i.$$

**C.1**

$\bar{Y}$  is called the **sample average** but, unlike in Appendix A where we defined the sample average of a set of numbers as a descriptive statistic,  $\bar{Y}$  is now viewed as an estimator. Given any outcome of the random variables  $Y_1, \dots, Y_n$ , we use the same rule to estimate  $\mu$ : we simply average them. For actual data outcomes  $\{y_1, \dots, y_n\}$ , the **estimate** is just the average in the sample:  $\bar{y} = (y_1 + y_2 + \dots + y_n)/n$ .

### Example C.1

#### [City Unemployment Rates]

Suppose we obtain the following sample of unemployment rates for 10 cities in the United States:

| City | Unemployment Rate |
|------|-------------------|
| 1    | 5.1               |
| 2    | 6.4               |
| 3    | 9.2               |
| 4    | 4.1               |
| 5    | 7.5               |
| 6    | 8.3               |
| 7    | 2.6               |
| 8    | 3.5               |
| 9    | 5.8               |
| 10   | 7.5               |

Our estimate of the average city unemployment rate in the United States is  $\bar{y} = 6.0$ . Each sample generally results in a different estimate. But the *rule* for obtaining the estimate is the same, regardless of which cities appear in the sample, or how many.

More generally, an estimator  $W$  of a parameter  $\theta$  can be expressed as an abstract mathematical formula:

$$W = h(Y_1, Y_2, \dots, Y_n), \quad \text{C.2}$$

for some known function  $h$  of the random variables  $Y_1, Y_2, \dots, Y_n$ . As with the special case of the sample average,  $W$  is a random variable because it depends on the random sample: as we obtain different random samples from the population, the value of  $W$  can change. When a particular set of numbers, say,  $\{y_1, y_2, \dots, y_n\}$ , is plugged into the function  $h$ , we obtain an *estimate* of  $\theta$ , denoted  $w = h(y_1, \dots, y_n)$ . Sometimes,  $W$  is called a point estimator and  $w$  a point estimate to distinguish these from *interval* estimators and estimates, which we will come to in Section C.5.

For evaluating estimation procedures, we study various properties of the probability distribution of the random variable  $W$ . The distribution of an estimator is often called its **sampling distribution**, because this distribution describes the likelihood of various outcomes of  $W$  across different random samples. Because there are unlimited rules for combining data to estimate parameters, we need some sensible criteria for choosing among estimators, or at least for eliminating some estimators from consideration. Therefore, we must leave the realm of descriptive statistics, where we compute things such as sample average to simply summarize a body of data. In mathematical statistics, we study the sampling distributions of estimators.

## Unbiasedness

In principle, the entire sampling distribution of  $W$  can be obtained given the probability distribution of  $Y_i$  and the function  $h$ . It is usually easier to focus on a few features of the distribution of  $W$  in evaluating it as an estimator of  $\theta$ . The first important property of an estimator involves its expected value.

**Unbiased Estimator.** An estimator,  $W$  of  $\theta$ , is an **unbiased estimator** if

$$E(W) = \theta, \quad \text{C.3}$$

for all possible values of  $\theta$ .

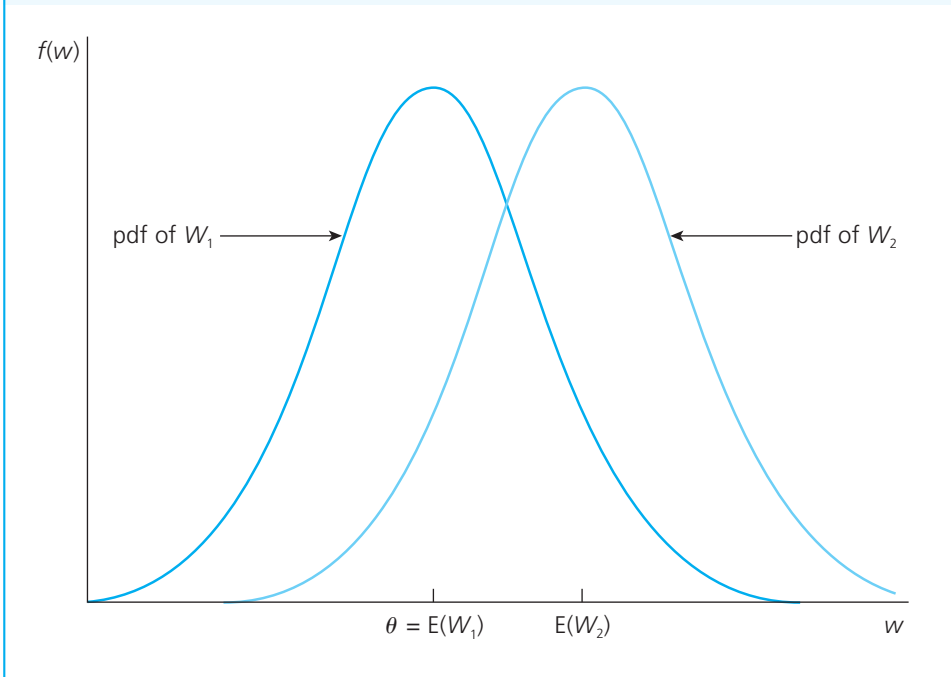
If an estimator is unbiased, then its probability distribution has an expected value equal to the parameter it is supposed to be estimating. Unbiasedness does *not* mean that the estimate we get with any particular sample is equal to  $\theta$ , or even very close to  $\theta$ . Rather, if we could *indefinitely* draw random samples on  $Y$  from the population, compute an estimate each time, and then average these estimates over all random samples, we would obtain  $\theta$ . This thought experiment is abstract because, in most applications, we just have one random sample to work with.

For an estimator that is not unbiased, we define its bias as follows.

**Bias of an Estimator.** If  $W$  is a **biased estimator** of  $\theta$ , its **bias** is defined as

$$\text{Bias}(W) \equiv E(W) - \theta. \quad \text{C.4}$$

Figure C.1 shows two estimators; the first one is unbiased, and the second one has a positive bias.

**FIGURE C.1****An unbiased estimator,  $W_1$ , and an estimator with positive bias,  $W_2$ .**

The unbiasedness of an estimator and the size of any possible bias depend on the distribution of  $Y$  and on the function  $h$ . The distribution of  $Y$  is usually beyond our control (although we often choose a *model* for this distribution): it may be determined by nature or social forces. But the choice of the rule  $h$  is ours, and if we want an unbiased estimator, then we must choose  $h$  accordingly.

Some estimators can be shown to be unbiased quite generally. We now show that the sample average  $\bar{Y}$  is an unbiased estimator of the population mean  $\mu$ , regardless of the underlying population distribution. We use the properties of expected values (E.1 and E.2) that we covered in Section B.3:

$$\begin{aligned} E(\bar{Y}) &= E\left((1/n) \sum_{i=1}^n Y_i\right) = (1/n)E\left(\sum_{i=1}^n Y_i\right) = (1/n)\left(\sum_{i=1}^n E(Y_i)\right) \\ &= (1/n)\left(\sum_{i=1}^n \mu\right) = (1/n)(n\mu) = \mu. \end{aligned}$$

For hypothesis testing, we will need to estimate the variance  $\sigma^2$  from a population with mean  $\mu$ . Letting  $\{Y_1, \dots, Y_n\}$  denote the random sample from the population with  $E(Y) = \mu$  and  $\text{Var}(Y) = \sigma^2$ , define the estimator as

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2,$$

**C.5**

which is usually called the **sample variance**. It can be shown that  $S^2$  is unbiased for  $\sigma^2$ :  $E(S^2) = \sigma^2$ . The division by  $n - 1$ , rather than  $n$ , accounts for the fact that the mean  $\mu$  is estimated rather than known. If  $\mu$  were known, an unbiased estimator of  $\sigma^2$  would be  $n^{-1} \sum_{i=1}^n (Y_i - \mu)^2$ , but  $\mu$  is rarely known in practice.

Although unbiasedness has a certain appeal as a property for an estimator—indeed, its antonym, “biased,” has decidedly negative connotations—it is not without its problems. One weakness of unbiasedness is that some reasonable, and even some very good, estimators are not unbiased. We will see an example shortly.

Another important weakness of unbiasedness is that unbiased estimators exist that are actually quite poor estimators. Consider estimating the mean  $\mu$  from a population. Rather than using the sample average  $\bar{Y}$  to estimate  $\mu$ , suppose that, after collecting a sample of size  $n$ , we discard all of the observations except the first. That is, our estimator of  $\mu$  is simply  $W \equiv Y_1$ . This estimator is unbiased because  $E(Y_1) = \mu$ . Hopefully, you sense that ignoring all but the first observation is not a prudent approach to estimation: it throws out most of the information in the sample. For example, with  $n = 100$ , we obtain 100 outcomes of the random variable  $Y$ , but then we use only the first of these to estimate  $E(Y)$ .

## The Sampling Variance of Estimators

The example at the end of the previous subsection shows that we need additional criteria to evaluate estimators. Unbiasedness only ensures that the sampling distribution of an estimator has a mean value equal to the parameter it is supposed to be estimating. This is fine, but we also need to know how spread out the distribution of an estimator is. An estimator can be equal to  $\theta$ , on average, but it can also be very far away with large probability. In Figure C.2,  $W_1$  and  $W_2$  are both unbiased estimators of  $\theta$ . But the distribution of  $W_1$  is more tightly centered about  $\theta$ : the probability that  $W_1$  is greater than any given distance from  $\theta$  is less than the probability that  $W_2$  is greater than that same distance from  $\theta$ . Using  $W_1$  as our estimator means that it is less likely that we will obtain a random sample that yields an estimate very far from  $\theta$ .

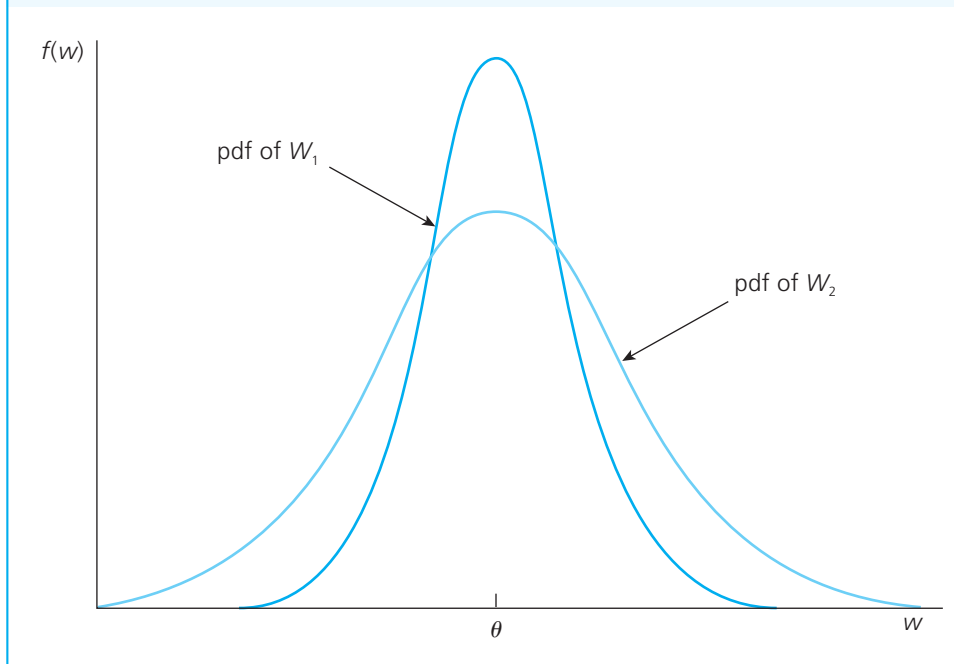
To summarize the situation shown in Figure C.2, we rely on the variance (or standard deviation) of an estimator. Recall that this gives a single measure of the dispersion in the distribution. The variance of an estimator is often called its **sampling variance** because it is the variance associated with a sampling distribution. Remember, the sampling variance is not a random variable; it is a constant, but it might be unknown.

We now obtain the variance of the sample average for estimating the mean  $\mu$  from a population:

$$\begin{aligned} \text{Var}(\bar{Y}) &= \text{Var}\left(\left(1/n\right)\sum_{i=1}^n Y_i\right) = (1/n^2)\text{Var}\left(\sum_{i=1}^n Y_i\right) = (1/n^2)\left(\sum_{i=1}^n \text{Var}(Y_i)\right) \\ &= (1/n^2)\left(\sum_{i=1}^n \sigma^2\right) = (1/n^2)(n\sigma^2) = \sigma^2/n. \end{aligned}$$

C.6

Notice how we used the properties of variance from Sections B.3 and B.4 (VAR.2 and VAR.4), as well as the independence of the  $Y_i$ . To summarize: If  $\{Y_i; i = 1, 2, \dots, n\}$  is a random sample from a population with mean  $\mu$  and variance  $\sigma^2$ , then  $\bar{Y}$  has the same mean as the population, but its sampling variance equals the population variance,  $\sigma^2$ , divided by the sample size.

**FIGURE C.2****The sampling distributions of two unbiased estimators of  $\theta$ .**

An important implication of  $\text{Var}(\bar{Y}) = \sigma^2/n$  is that it can be made very close to zero by increasing the sample size  $n$ . This is a key feature of a reasonable estimator, and we return to it in Section C.3.

As suggested by Figure C.2, among unbiased estimators, we prefer the estimator with the smallest variance. This allows us to eliminate certain estimators from consideration. For a random sample from a population with mean  $\mu$  and variance  $\sigma^2$ , we know that  $\bar{Y}$  is unbiased, and  $\text{Var}(\bar{Y}) = \sigma^2/n$ . What about the estimator  $Y_1$ , which is just the first observation drawn? Because  $Y_1$  is a random draw from the population,  $\text{Var}(Y_1) = \sigma^2$ . Thus, the difference between  $\text{Var}(Y_1)$  and  $\text{Var}(\bar{Y})$  can be large even for small sample sizes. If  $n = 10$ , then  $\text{Var}(Y_1)$  is 10 times as large as  $\text{Var}(\bar{Y}) = \sigma^2/10$ . This gives us a formal way of excluding  $Y_1$  as an estimator of  $\mu$ .

To emphasize this point, Table C.1 contains the outcome of a small simulation study. Using the statistical package Stata®, 20 random samples of size 10 were generated from a normal distribution, with  $\mu = 2$  and  $\sigma^2 = 1$ ; we are interested in estimating  $\mu$  here. For each of the 20 random samples, we compute two estimates,  $y_1$  and  $\bar{y}$ ; these values are listed in Table C.1. As can be seen from the table, the values for  $y_1$  are much more spread out than those for  $\bar{y}$ :  $y_1$  ranges from  $-0.64$  to  $4.27$ , while  $\bar{y}$  ranges only from  $1.16$  to  $2.58$ . Further, in 16 out of 20 cases,  $\bar{y}$  is closer than  $y_1$  to  $\mu = 2$ . The average of  $y_1$  across the simulations is about 1.89, while that for  $\bar{y}$  is 1.96. The fact that these averages are close to 2 illustrates the unbiasedness of both estimators (and we could get these averages closer to 2 by doing more than 20 replications). But comparing just the average outcomes across random draws masks the fact that the sample average  $\bar{Y}$  is far superior to  $Y_1$  as an estimator of  $\mu$ .

**TABLE C.1****Simulation of Estimators for a Normal( $\mu, 1$ ) Distribution with  $\mu = 2$** 

| Replication | $y_1$ | $\bar{y}$ |
|-------------|-------|-----------|
| 1           | -0.64 | 1.98      |
| 2           | 1.06  | 1.43      |
| 3           | 4.27  | 1.65      |
| 4           | 1.03  | 1.88      |
| 5           | 3.16  | 2.34      |
| 6           | 2.77  | 2.58      |
| 7           | 1.68  | 1.58      |
| 8           | 2.98  | 2.23      |
| 9           | 2.25  | 1.96      |
| 10          | 2.04  | 2.11      |
| 11          | 0.95  | 2.15      |
| 12          | 1.36  | 1.93      |
| 13          | 2.62  | 2.02      |
| 14          | 2.97  | 2.10      |
| 15          | 1.93  | 2.18      |
| 16          | 1.14  | 2.10      |
| 17          | 2.08  | 1.94      |
| 18          | 1.52  | 2.21      |
| 19          | 1.33  | 1.16      |
| 20          | 1.21  | 1.75      |

## Efficiency

Comparing the variances of  $\bar{Y}$  and  $Y_1$  in the previous subsection is an example of a general approach to comparing different unbiased estimators.

**Relative Efficiency.** If  $W_1$  and  $W_2$  are two unbiased estimators of  $\theta$ ,  $W_1$  is efficient relative to  $W_2$  when  $\text{Var}(W_1) \leq \text{Var}(W_2)$  for all  $\theta$ , with strict inequality for at least one value of  $\theta$ .

Earlier, we showed that, for estimating the population mean  $\mu$ ,  $\text{Var}(\bar{Y}) < \text{Var}(Y_1)$  for any value of  $\sigma^2$  whenever  $n > 1$ . Thus,  $\bar{Y}$  is efficient relative to  $Y_1$  for estimating  $\mu$ . We cannot always choose between unbiased estimators based on the smallest variance criterion: given two unbiased estimators of  $\theta$ , one can have smaller variance from some values of  $\theta$ , while the other can have smaller variance for other values of  $\theta$ .

If we restrict our attention to a certain class of estimators, we can show that the sample average has the smallest variance. Problem C.2 asks you to show that  $\bar{Y}$  has the smallest variance among all unbiased estimators that are also linear functions of  $Y_1, Y_2, \dots, Y_n$ . The assumptions are that the  $Y_i$  have common mean and variance, and that they are pairwise uncorrelated.

If we do not restrict our attention to unbiased estimators, then comparing variances is meaningless. For example, when estimating the population mean  $\mu$ , we can use a trivial estimator that is equal to zero, regardless of the sample that we draw. Naturally, the variance of this estimator is zero (since it is the same value for every random sample). But the bias of this estimator is  $-\mu$ , so it is a very poor estimator when  $|\mu|$  is large.

One way to compare estimators that are not necessarily unbiased is to compute the **mean squared error (MSE)** of the estimators. If  $W$  is an estimator of  $\theta$ , then the MSE of  $W$  is defined as  $\text{MSE}(W) = E[(W - \theta)^2]$ . The MSE measures how far, on average, the estimator is away from  $\theta$ . It can be shown that  $\text{MSE}(W) = \text{Var}(W) + [\text{Bias}(W)]^2$ , so that  $\text{MSE}(W)$  depends on the variance and bias (if any is present). This allows us to compare two estimators when one or both are biased.

## C.3 Asymptotic or Larger Sample Properties of Estimators

In Section C.2, we encountered the estimator  $Y_1$  for the population mean  $\mu$ , and we saw that, even though it is unbiased, it is a poor estimator because its variance can be much larger than that of the sample mean. One notable feature of  $Y_1$  is that it has the same variance for any sample size. It seems reasonable to require any estimation procedure to improve as the sample size increases. For estimating a population mean  $\mu$ ,  $\bar{Y}$  improves in the sense that its variance gets smaller as  $n$  gets larger;  $Y_1$  does not improve in this sense.

We can rule out certain silly estimators by studying the *asymptotic* or *large sample* properties of estimators. In addition, we can say something positive about estimators that are not unbiased and whose variances are not easily found.

Asymptotic analysis involves approximating the features of the sampling distribution of an estimator. These approximations depend on the size of the sample. Unfortunately, we are necessarily limited in what we can say about how “large” a sample size is needed for asymptotic analysis to be appropriate; this depends on the underlying population distribution. But large sample approximations have been known to work well for sample sizes as small as  $n = 20$ .

### Consistency

The first asymptotic property of estimators concerns how far the estimator is likely to be from the parameter it is supposed to be estimating as we let the sample size increase indefinitely.